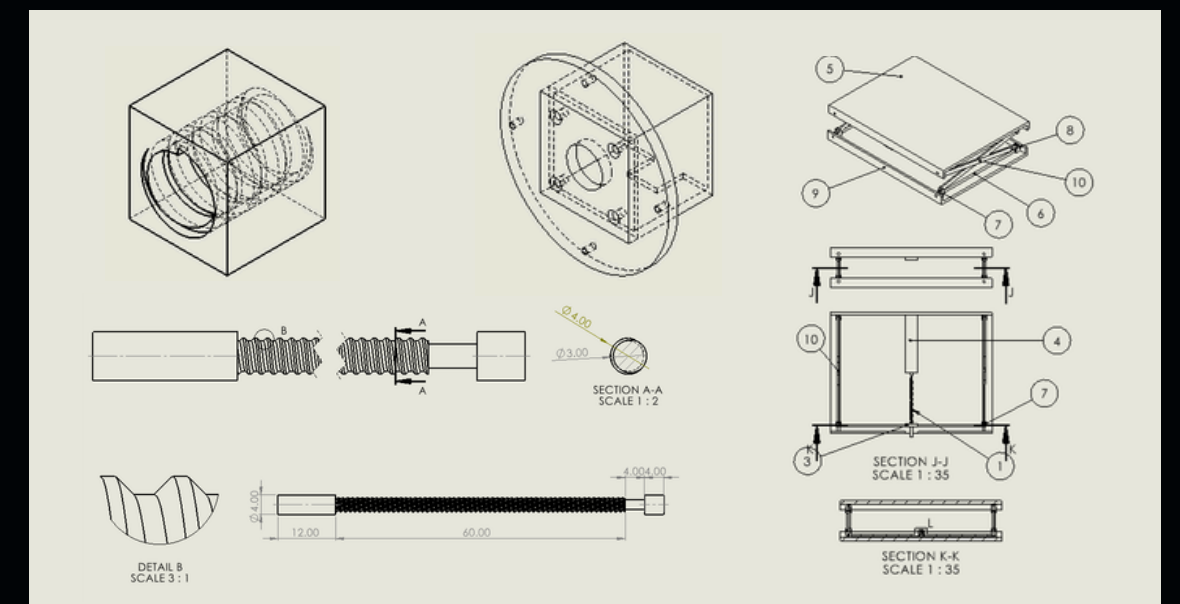
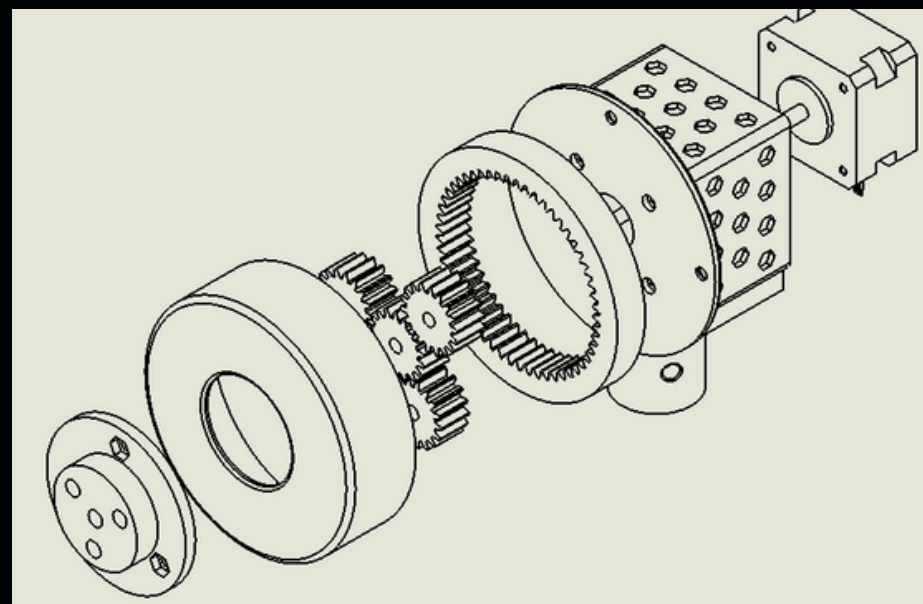


# BROOKLYN CLAEYS

MECHANICAL ENGINEERING | BIOMEDICAL MINOR  
University of Calgary

MECHANICAL SYSTEMS R&D |  
MECHATRONICS • DYNAMICS • DESIGN

ENGINEERING PORTFOLIO



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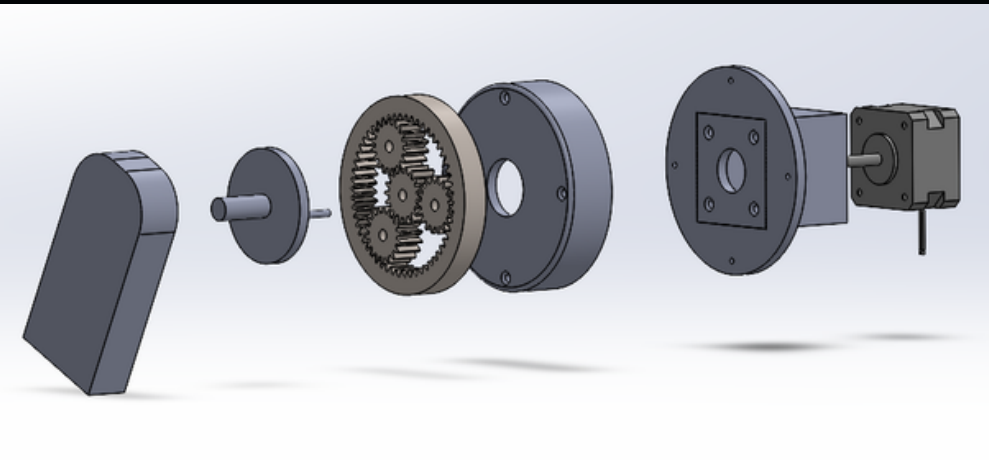
# Yankauer Robotic Arm

A BMERIT Club project developed with Foothills Hospital to create an eye-tracking-controlled robotic arm that positions a Yankauer suction device for patients with spinal cord injuries. I designed and built the shoulder joint, then became Mechanical Project Lead for the current full-arm redesign.



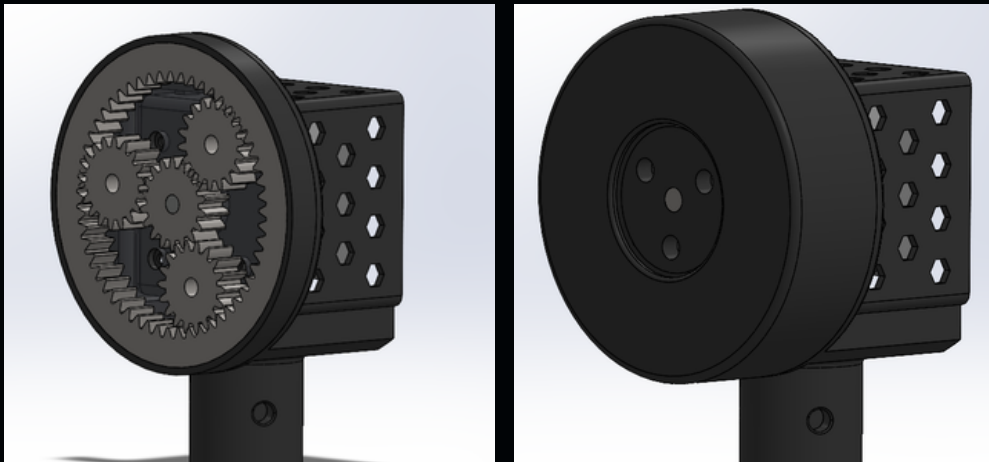
## Shoulder Joint Redesign

I inherited the team’s existing shoulder design with exposed gearing and redesigned it as a compact, enclosed planetary joint. I designed, printed, and assembled the complete mechanism, integrating the motor and arm attachments and adding passive cooling vents to the housing. The gears fit on the first build and the joint moved successfully during its first powered test.



## Prototype Revisions

Testing exposed weak load-bearing details in the first joint. I replaced the printed planet supports with metal fasteners and redesigned the output from a printed attachment to a three-screw connection, then revised spacing and retention during arm integration.

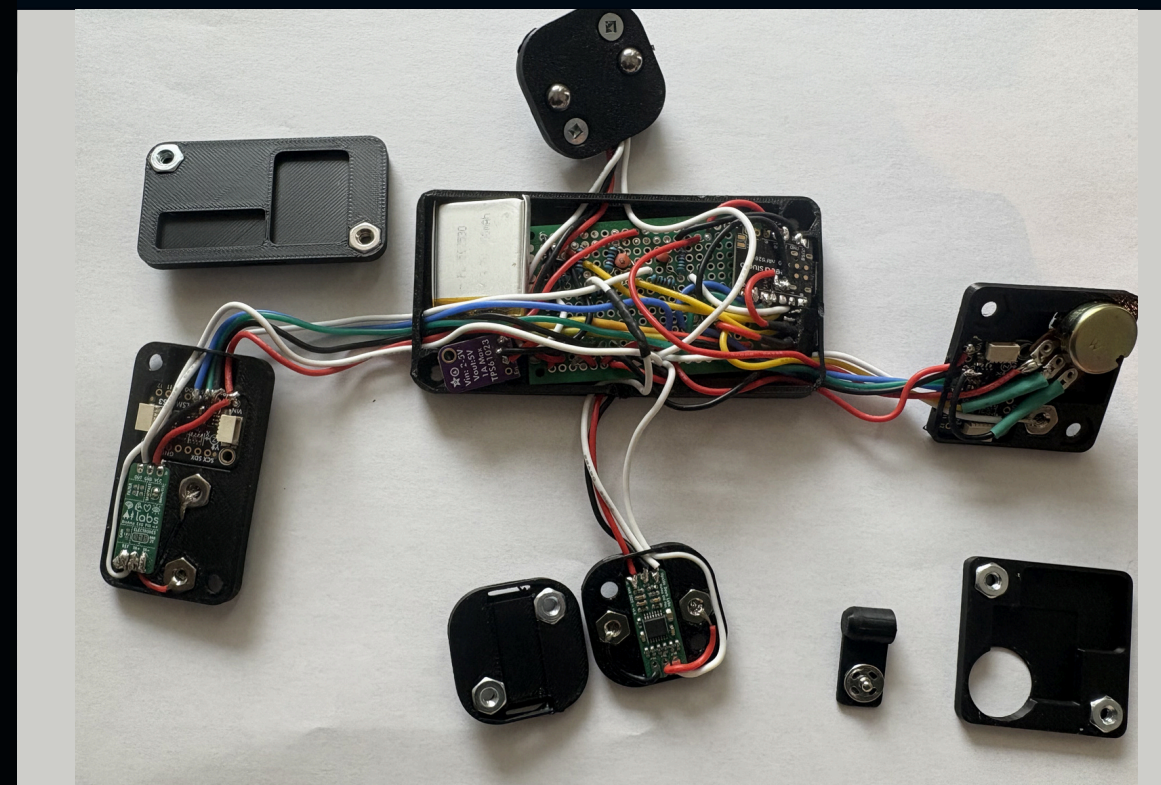


## Current Full Arm Redesign

Following feedback from Foothills, I am redesigning the full arm around the planetary-joint approach, replacing the previous linear-actuator layout with a free-standing system of powered rotary joints. The next joint revision adds output bearings for better support under load and lower friction.

# Wearable EMG Sleeve

A wearable sleeve that combines EMG sensing with measured arm motion to provide live feedback on muscle activation, joint movement, repetition timing, and form during exercise. It uses three EMG sensors, two IMUs, and an elbow potentiometer, and currently works with biceps curls, triceps extensions, and lateral raises, sending the combined data to a Bluetooth dashboard.



## System Overview

- Removable EMG pods + elbow control module
- 3 EMG channels, 2 IMUs + elbow-angle sensing
- XIAO nRF52840 Sense + BLE dashboard

## Integration + Testing

I designed the removable pods, electrode placement, elbow module and cable routing to keep the sensing locations fixed while preserving access to the electronics. I integrated the sensors, controller and dashboard to combine EMG, orientation and elbow-angle data during exercise. During testing, pod pressure, electrode contact and cable movement changed EMG quality, so the mechanical packaging and sensor setup were refined together.

### Four-Body Model + Experimental Inputs

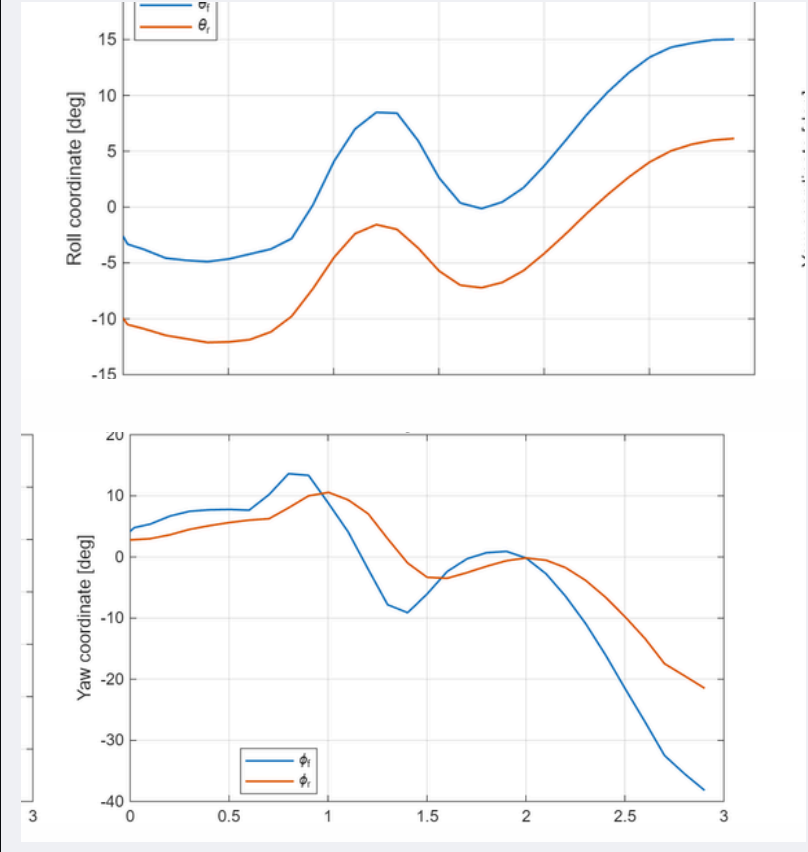
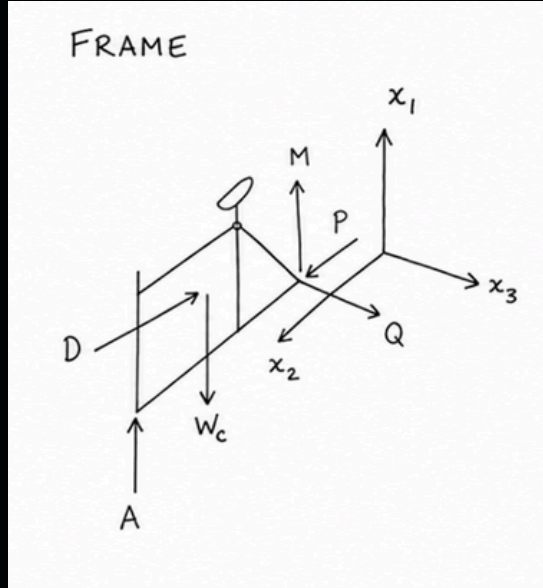
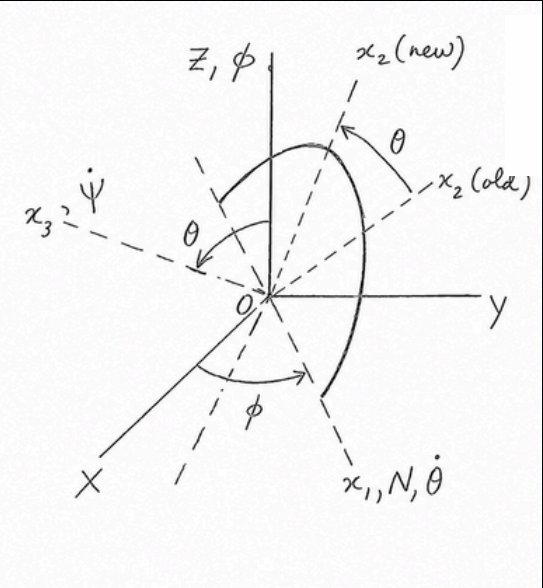
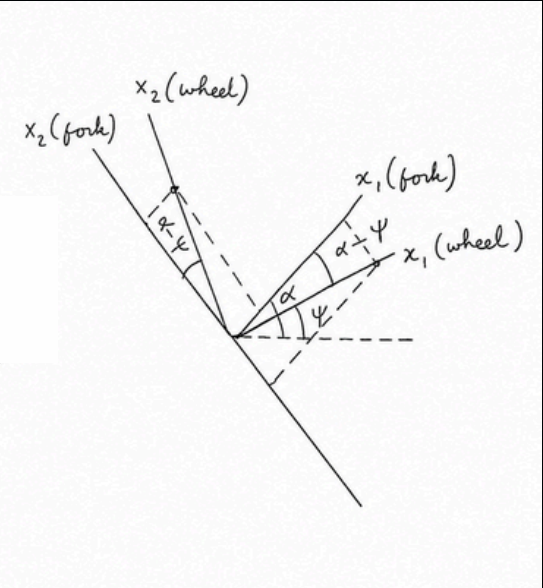
Front and rear IMU data, wheel-speed measurements, and 16 instrumented riding trials were mapped into a four-body bicycle model. I processed the measured motion, mapped SolidWorks geometry and mass properties into the analytical coordinate system, and updated the frame and rider representation used in the dynamics model.

### Multibody Kinematics + Coupled Dynamics

The wheel, fork and frame move in different coordinate systems, so their motion had to be transformed into a common formulation. I independently derived the coordinate transforms, checked them against Dr. Budiman’s geometric method, and carried them into the coupled Newton–Euler force, moment and constraint equations.

### MATLAB Implementation + Model Diagnostics

I developed and continue to revise the MATLAB implementation, including experimental data processing, inverse dynamics, batch analysis and model diagnostics. I built checks for residuals, rank and conditioning to separate modelling or numerical issues from actual bicycle behaviour. A forward-dynamics model is in development, with current work focused on debugging the equations, parameters and signal processing.



$$R_{\text{fork} \rightarrow \text{frame}} = R_{\alpha} R_{\psi} R_{\phi}$$

$$R_{\text{fork} \rightarrow \text{frame}} = \begin{bmatrix} \cos \alpha_e & \sin \alpha_e & 0 \\ -\sin \alpha_e & \cos \alpha_e & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha_k & -\sin \alpha_k & 0 \\ \sin \alpha_k & \cos \alpha_k & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi_k & -\sin \phi_k \\ 0 & \sin \phi_k & \cos \phi_k \end{bmatrix}$$

$$R_{\text{fork} \rightarrow \text{frame}} = \begin{bmatrix} \cos(\alpha_e - \alpha_k) & \sin(\alpha_e - \alpha_k) \cos \phi_k & -\sin(\alpha_e - \alpha_k) \sin \phi_k \\ -\sin(\alpha_e - \alpha_k) & \cos(\alpha_e - \alpha_k) \cos \phi_k & -\cos(\alpha_e - \alpha_k) \sin \phi_k \\ 0 & \sin \phi_k & \cos \phi_k \end{bmatrix}$$

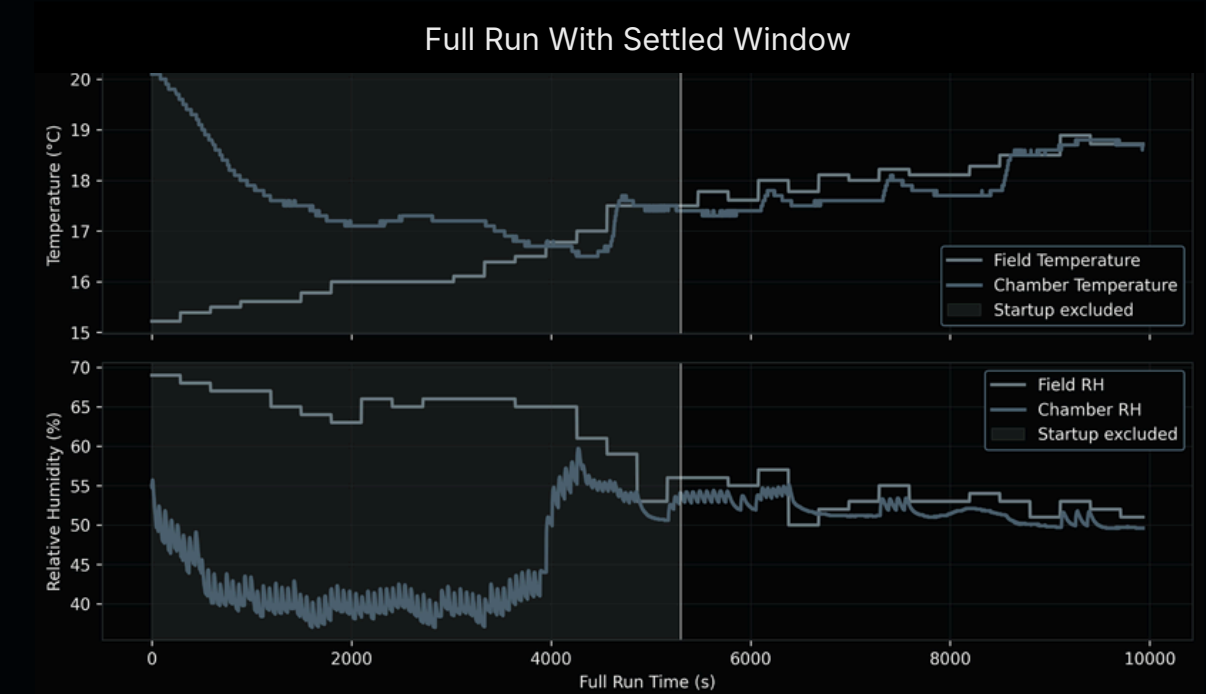
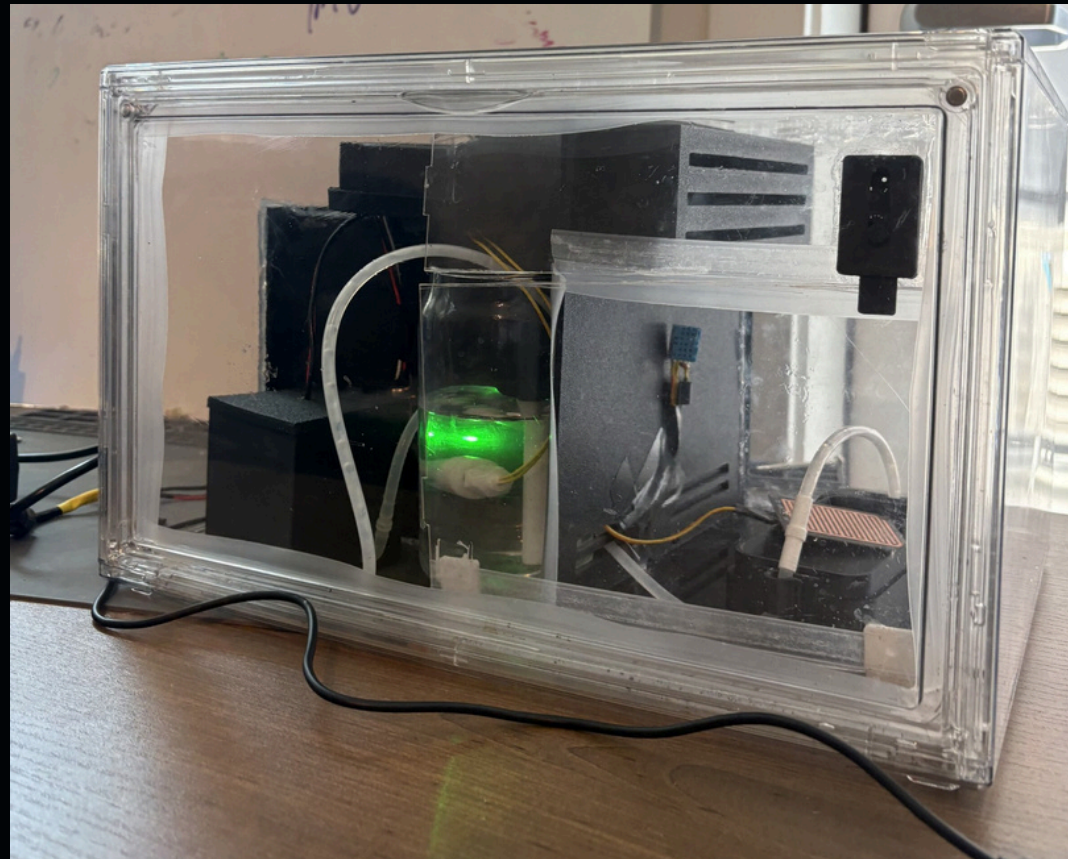
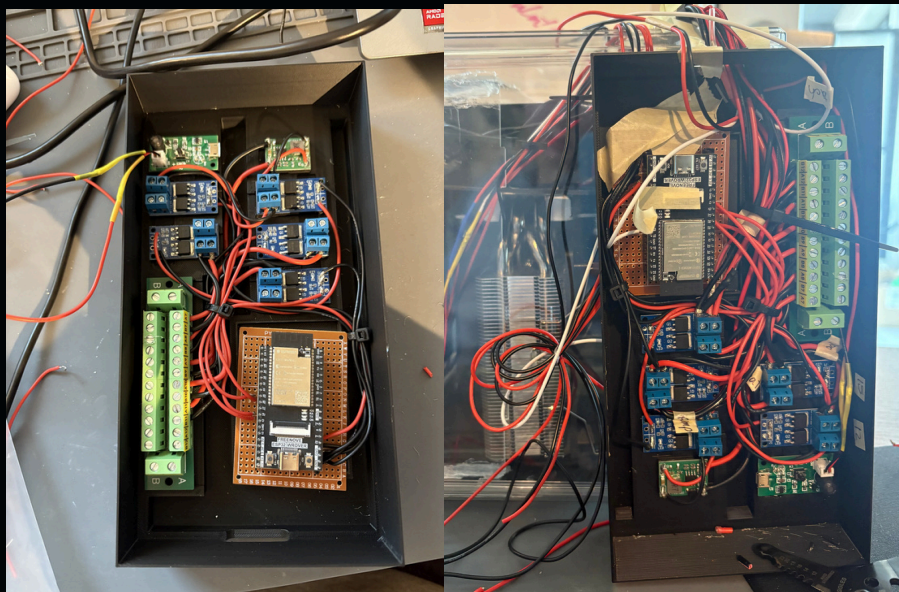
# Environmental Control Chamber

2026 • Personal Project

I built an ESP32-controlled environmental chamber to reproduce changing field temperature and humidity conditions for weather-sensitive crop testing. Heating, cooling, humidification, airflow and water delivery respond to manual or live weather-station targets.

## Controls + Electrical

- Ecowitt API + manual target inputs
- ESP32 sensor feedback + control logic
- MOSFET-switched 12 V actuator branches
- Float-switch + sensor-fault interlocks
- Live dashboard + CSV test logging



## System Design

A dual-wall enclosure limits heat transfer, while an internal plenum mixes conditioned air before it enters the chamber. Vent placement directs airflow through the upper test volume, with heating, cooling, humidification and water delivery integrated around the same airflow path.

## Results

- 77.2 min settled evaluation window
- 0.28 °C temperature MAE
- 1.94 percentage points RH MAE